2.1 Babylonian Numeration

The mathematics of Mesopotamia from 2500 BCE[1] to 300 BCE is usually called "Babylonian". This oversimplifies the history of that region. During that very long period of time, a variety of civilizations occupied that area. The Sumerians first invented a unique method of writing sometime during the third millennium BCE, and successive civilizations like the Babylonians adopted it.

They didn't use paper, but instead used a small stylus tool to put marks into clay tablets. The clay[2] pieces were often small enough to fit in one hand. The stick made two different wedge-shaped marks 𒁹 or 𒁺 when pressed into the clay in two different ways.[3]

The writing is sometimes called *cuneiform* ("kyoo-nee-uh-form") which means "wedge-shaped".

[1] Before Common Era

[2] clay : *argile*

[3] Writing cuneiform https://youtu.be/HbZ2asfyHcA

On the next page you will find a recreation of a real tablet, presented close to its actual size. It was discovered in the Sumerian city of Nippur (in modern-day Iraq), and dates to around 1500 BCE. We're not completely sure what this is, but most scholars suspect that it is a **homework exercise**, preserved for thousands of years. When the sun hardened the clay, these tablets became permanent records. Thousands of them still exist in museums around the world.

Of course, it's not preserved perfectly, and dealing with the parts that have become damaged is part of the challenge (and part of the fun) of working with old texts like this. If you study the picture closely, you may discover that you can figure out a lot about Babylonian numerals.

Meet a Babylonian Tablet

Study the image below an try to answer

1. How do Babylonian numerals work?[1]
2. Describe the mathematics on this tablet
3. Write the number 72 in Babylonian numerals.
4. Why can this tablet remind us of a school homework?

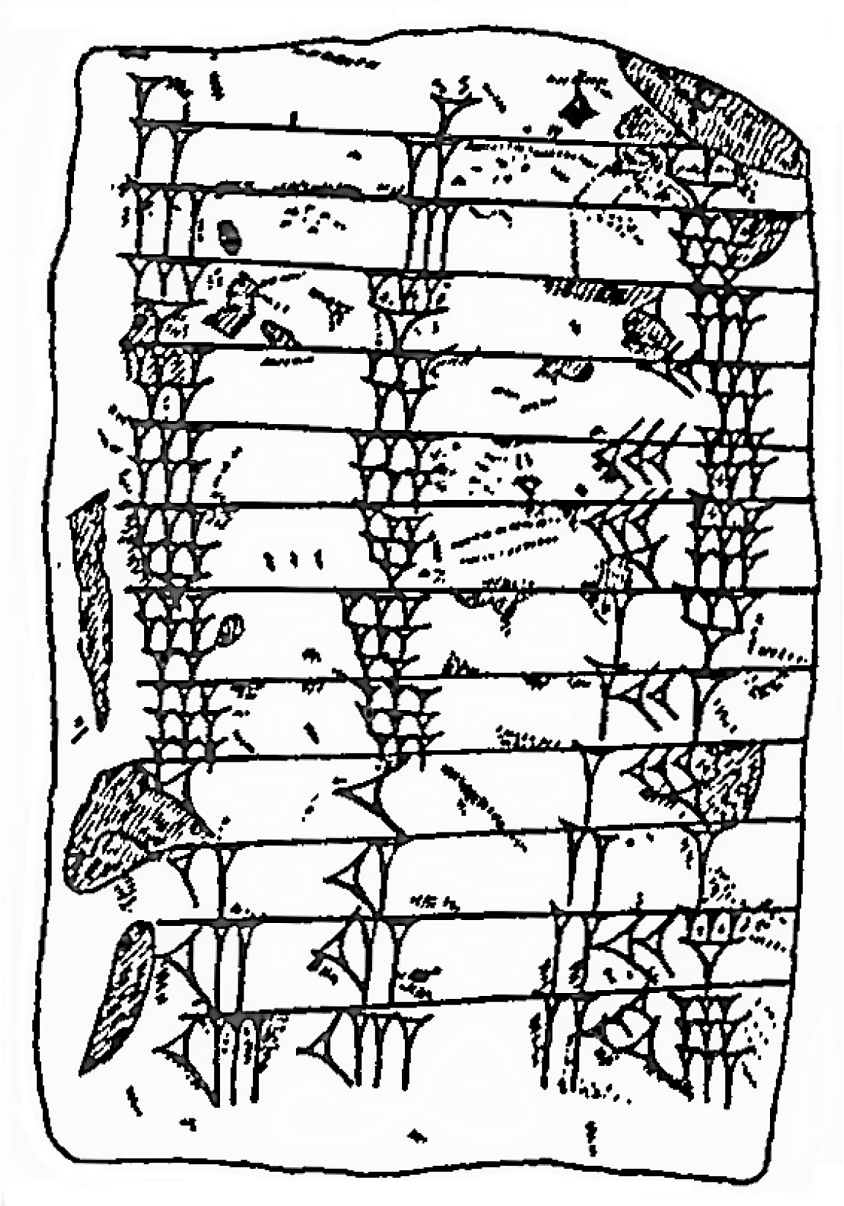

Figure 2.1 – This is image 26 on Plate 16 in Hermannn Vollrat Hilprecht, The Babylonian Expedition of the University of Pennsylvania. 1906

[1]symbols and can be written and